

Guide 45 — Water and Wastewater Treatment Plant Operator

Controlled English working master — version 2.0

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What this job is

Water and wastewater treatment plant and system operators help keep drinking-water and wastewater systems operating within approved public-health, environmental, safety, and plant requirements. The work combines process monitoring, equipment observation, sampling and records, routine maintenance support, alarms and communications, computerized controls, housekeeping, and carefully controlled operating decisions.

Employers may use titles such as water treatment operator, wastewater treatment operator, plant operator, treatment-system operator, operator-in-training, utility operator, distribution or collection operator, process operator, or treatment technician. Similar titles can carry very different duties, certification levels, facility classifications, shifts, and legal responsibilities.

The job title alone does not authorize a worker to change process settings, select or dose chemicals, enter a confined space, perform electrical work, apply lockout/tagout, bypass treatment, release laboratory results, make regulatory compliance determinations, or act as an operator in responsible charge. Those duties depend on the jurisdiction, facility class, operator level, employer authorization, approved procedures, current training, and qualified supervision.

This guide supports career exploration. It is not a plant operating manual, sampling protocol, laboratory method, chemical-dosing instruction, confined-space program, energy-control procedure, emergency plan, permit interpretation, or authorization to operate independently.

What operators actually do

Depending on the plant, shift, certification, training, and authorization, work may include:

- review shift logs, work orders, alarms, permits, approved procedures, safety information, and operating records;
- observe tanks, basins, pumps, valves, piping, meters, gauges, treatment units, filters, clarifiers, blowers, disinfection systems, chemical systems, and support equipment;
- monitor approved process indicators and computerized control or SCADA displays within assigned authority;
- record readings, samples, inspections, maintenance observations, alarms, corrective actions, and shift handovers;
- collect or assist with samples and routine tests under the facility's approved methods and supervision;
- perform or assist with authorized cleaning, housekeeping, inspection, preventive maintenance, and minor equipment tasks;

- receive, stage, or document materials and chemicals within employer procedures and assigned role;
- communicate abnormal readings, leaks, odors, alarms, spills, equipment problems, public-health concerns, and environmental concerns;
- coordinate with certified operators, supervisors, laboratory staff, mechanics, electricians, engineers, environmental staff, contractors, regulators, emergency personnel, and other utility workers; and
- stop and escalate when a condition, reading, task, atmosphere, energy state, sample, process change, or authorization is uncertain.

BLS and O*NET describe broad occupational tasks. They do not establish that every trainee or entry-level worker may perform every task. An operator-in-training or new employee may begin with observation, readings, documentation, housekeeping, supervised sampling, and simple maintenance support before progressing through the employer's and jurisdiction's qualification process.

Work environment and schedules

Water and wastewater facilities can operate continuously. BLS notes that operators may work rotating shifts, nights, weekends, holidays, and emergency callouts. Some facilities are staffed around the clock; others use on-call arrangements or smaller teams.

Work may occur indoors and outdoors around wet surfaces, tanks, basins, pits, manholes, pumps, motors, piping, chemical areas, laboratories, vehicles, ladders, platforms, and computerized control rooms. Conditions can include noise, odors, heat, cold, rain, poor footing, sewage or sludge, aerosols, corrosive or toxic substances, dangerous gases, confined spaces, moving equipment, pressure, and difficult-to-access locations.

Read each job posting carefully. A trainee role at a small drinking-water plant can differ significantly from a certified wastewater operator role at a large facility, a collection-system role, or a position with responsible-charge duties.

Physical demands

The work may involve standing, walking, climbing stairs or ladders, bending, kneeling, carrying tools or samples, opening approved access points, reading instruments, working around equipment, and responding to alarms or emergencies. Employers may set lawful job-related fitness, driving, PPE, respirator, vaccination, medical-surveillance, or emergency-response requirements based on the actual hazards and essential functions.

Do not treat physical demands as a test of toughness. Ask how the employer controls lifting, slips, falls, traffic, heat, fatigue, exposure, confined spaces, rescue, lone work, and emergency response, and how lawful accommodations are handled for the actual job.

Safety comes before production

Water and wastewater work can expose people to pathogens, sharps, sewage and sludge, aerosols, treatment chemicals, incompatible substances, chlorine or other toxic gases, hydrogen

sulfide, oxygen-deficient atmospheres, confined spaces, falls and drowning, electricity, stored energy, pressure, rotating machinery, vehicles, slips, lifting, heat, and fatigue.

Do not improvise chemical dosing or mixing, confined-space entry or rescue, atmospheric testing, lockout/tagout, energized electrical work, line opening, pressure release, bypassing, disinfection, laboratory release, process-control changes, spill response, or return to service. Use only the employer's approved procedures, trained roles, permits, PPE, monitoring, communications, and supervision.

Stop work and escalate when:

- a task is outside your certification, training, assigned role, or employer authorization;
- a process reading, alarm, sample result, or SCADA indication is unexplained or inconsistent;
- a chemical container, line, transfer, label, safety data sheet, concentration, compatibility, or authorization is unclear;
- there is a chemical leak, unusual odor, visible vapor, spill, damaged container, or suspected incompatible substance;
- a gas alarm sounds, a person has symptoms, ventilation is uncertain, or an atmosphere may be oxygen-deficient or toxic;
- a pit, manhole, vault, tank, wet well, sewer, basin, or other space may be a permit-required confined space and the site program has not authorized the work;
- sewage, sludge, sharps, blood, aerosols, or other biological exposure is uncontrolled or required hygiene/PPE is unavailable;
- electrical condition, hazardous energy, pressure, line contents, guards, rotation, lockout status, or equipment identity is uncertain;
- a pump, motor, valve, blower, mixer, conveyor, vehicle, platform, ladder, railing, cover, or lifting device appears damaged or abnormal;
- there is a fall or drowning hazard without the required controls;
- a bypass, discharge, treatment interruption, laboratory decision, public-health issue, or regulatory reporting question is outside your authority;
- communications, supervision, rescue readiness, emergency response, or required second-person coverage is missing; or
- production pressure conflicts with approved safety, environmental, or public-health controls.

Confined spaces, hydrogen sulfide, and toxic atmospheres

OSHA identifies sewer and wastewater treatment as settings where hydrogen sulfide may occur. It can collect in low or enclosed areas such as pits, manholes, sewers, and vaults. Smell is not a reliable warning.

A career guide cannot determine whether a space is permit-required, select an atmospheric monitor, establish acceptable readings, authorize entry, set ventilation, assign an attendant, or create a rescue plan. Follow the employer's confined-space program and only perform the role for which you are trained, equipped, and authorized.

Never enter an uncertain space or attempt an improvised rescue. If an alarm occurs, communication is lost, ventilation fails, a person becomes ill, or the atmosphere or space classification is uncertain, leave or remain out as required by the site procedure and contact the responsible trained personnel.

Chemicals and biological exposure

Treatment facilities may use disinfectants, coagulants, acids, bases, polymers, oxidizers, or other process chemicals. Wastewater facilities may also involve sewage, sludge, sharps, aerosols, and biological hazards.

Do not select chemicals, calculate or set doses, mix incompatible substances, transfer material, neutralize a spill, choose PPE, or perform decontamination from instructions in a career guide or AI response. Use the plant's approved procedure, labels, safety data sheets, exposure-control program, qualified personnel, and emergency process.

For biological exposure, follow the employer's hygiene, PPE, sharps, vaccination or medical provisions, exposure reporting, and decontamination requirements. Report needlesticks, splash exposures, symptoms, damaged PPE, or uncontrolled sewage contact immediately through the facility's process.

Electrical, mechanical, pressure, and hazardous-energy boundaries

Pumps, motors, blowers, mixers, valves, conveyors, electrical panels, generators, chemical-feed equipment, compressed systems, and pressurized lines can retain or introduce hazardous energy even when equipment appears stopped.

OSHA's lockout/tagout framework assigns specific responsibilities to authorized and affected employees. Do not copy an isolation sequence, valve order, test method, lock placement, pressure-release step, electrical check, bypass, jumper, or restart procedure from this guide.

If equipment identity, energy state, pressure, line contents, guarding, remote-control status, or authorization is unclear, stop and contact the responsible authorized or qualified person. Never assume silence, zero flow on one indicator, or lack of visible movement proves equipment is safe.

Sampling, laboratory work, records, and compliance boundaries

Sampling and laboratory results can influence treatment decisions, public-health reporting, permit compliance, and environmental actions. Methods, preservation, chain of custody, calibration, acceptance criteria, release authority, and reporting vary by facility and jurisdiction.

A trainee may assist with routine sampling or testing under approved procedures, but this guide does not provide a test method or authorize interpretation. Do not change a process, release a result, certify compliance, alter a record, or decide that a discharge or drinking-water condition is acceptable unless that responsibility is explicitly within your authorized role.

Accurate records matter. Record only what you actually observed or performed, use the required units and timestamps, preserve corrections according to facility procedure, and escalate discrepancies rather than guessing.

SCADA, alarms, cybersecurity, and privacy

Many plants use computerized controls or SCADA systems to display process conditions and alarms. Access rights and operating authority differ by role.

Never share credentials, bypass access controls, use unauthorized remote-access tools, disable alarms, change setpoints, or copy sensitive plant information into public AI systems. Protect facility diagrams, treatment settings, chemical inventories, security information, customer records, laboratory results, incident details, public-health data, environmental reporting information, access routes, and photographs according to employer policy and law.

Treat unexpected SCADA behavior, unfamiliar prompts, unexplained remote sessions, configuration changes, missing data, or suspicious alarms as issues to verify through the approved operational and cybersecurity process.

Skills employers value

Employers commonly value:

- careful reading of procedures, logs, labels, safety information, and work orders;
- accurate observation, measurement, documentation, and shift handover;
- basic math, chemistry, biology, mechanical, electrical, and process awareness appropriate to the role;
- safe sampling and housekeeping habits within approved procedures;
- reliable communication and willingness to escalate uncertainty;
- comfort with computers, alarms, records, and controlled SCADA interfaces;
- attention to hygiene, PPE, equipment condition, slips, traffic, and site security;
- teamwork with certified operators, laboratory staff, maintenance, engineers, and supervisors; and
- disciplined respect for certification, facility classification, authorization, and regulated scope.

Do not describe yourself as a certified operator, operator in responsible charge, laboratory analyst, electrician, confined-space entrant or rescuer, authorized lockout employee, environmental compliance specialist, or qualified chemical handler unless you actually hold the required credential or employer authorization.

Education and entry pathways

United States

BLS classifies Water and Wastewater Treatment Plant and System Operators under SOC **51-8031**. The typical entry-level education is a **high school diploma or equivalent**; related work experience is typically **none**; and typical training is **long-term on-the-job training**.

BLS states that a state license is usually required, although entry-level workers often complete licensing requirements while receiving on-the-job training. Requirements and operator levels vary widely by state and facility.

A practical sequence is:

1. Compare current operator-in-training, trainee, utility operator, water operator, and wastewater operator postings in the location where you intend to work.
2. Identify the controlling state drinking-water and wastewater operator-certification authorities and the facility classification tied to the job.
3. Record required education, exams, experience, operator-in-training status, reciprocity, renewal, continuing education, driving, shift, and physical requirements.
4. Compare employer-paid training, Registered Apprenticeship, public community or technical college programs, and eligible aid before paying for a private course.
5. Build supervised evidence in documentation, math, basic science, maintenance support, safe sampling, computer use, and utility operations without working beyond authorization.
6. Progress only through the jurisdiction's certification rules and the employer's task-qualification process.

O*NET identifies **51-8031.00 Water and Wastewater Treatment Plant and System Operators** and lists apprenticeship titles including Waste Treatment Operator, Wastewater Systems Operator, Water Systems Operation Specialist, and Water Treatment Plant Operator. These titles are pathway evidence, not proof of a current opening or license.

Canada

Government of Canada Job Bank maps Wastewater Treatment Plant Operator to **Water and waste treatment plant operators, NOC 92101**.

Job Bank states that secondary school is required; college or industry courses in chemistry, pollution control, or related subjects are usually required; on-the-job training is provided; and previous field experience may be required. It also states that certification for water treatment/distribution and wastewater collection/treatment is required at levels related to plant classification.

Canada does not have one national operator license that overrides provincial or territorial requirements. Health Canada notes that provinces and territories are primarily responsible for drinking-water safety, while municipalities usually handle treatment and distribution. Wastewater facilities may also be subject to federal Wastewater Systems Effluent Regulations.

Verify the province or territory, facility class, operator class, operator-in-training rules, education, experience, examination, renewal, continuing education, reciprocity, and responsible-charge requirements before relying on a course or prior credential.

Colombia

SENA's Betowa catalog identifies related pathways including:

- **Operación de sistemas de potabilización de agua — Técnico, 12 months**
- **Saneamiento y salud ambiental — Técnico, 2,208 hours**
- **Análisis de riesgos en sistemas de agua y saneamiento — complementary training, 48 hours**

Availability is dynamic. A program profile may exist while no current cohort or places are open. Verify the live listing, municipality, modality, dates, admission requirements, and availability before planning around it.

The Ministry of Housing's published RAS Title C states that personnel operating potable-water treatment systems must have competency appropriate to their work and describes certification through SENA or another legally accredited personnel-certification body. Resolution 2115 of 2007 also references certified plant workers under the applicable national framework.

These sources do not authorize a person to operate every plant or perform every task. Current Ministry, SENA, ONAC, employer, service-provider, technical-standard, and local requirements must be checked because rules, standards, competency schemes, and accreditation can change.

Latin America and the Caribbean

Training, certification, and operator authority differ by country. OIT/Cinterfor's institutional network and REDIFP resources can help locate national vocational-training and competency-certification institutions.

Use the regional network as a locator, then verify the program, drinking-water authority, environmental authority, labor/safety authority, facility classification, operator requirements, and employer rules in the country where you intend to work. Do not assume that a course, certificate, operator level, or license transfers across borders.

Apprenticeships and employer training

In the United States, O*NET and Apprenticeship.gov support Registered Apprenticeship as a possible route under several water and wastewater operator titles. Use Apprenticeship.gov as a live locator and verify the sponsor, occupation, location, paid status, eligibility, registration, current opening, related instruction, and relationship to state certification.

Employer training may include site orientation, process fundamentals, sampling, laboratory support, equipment rounds, computerized controls, maintenance support, hazard communication, biological exposure controls, energy-control awareness, confined-space roles, PPE, emergency response, and supervised task sign-offs. Ask which training is paid, what certifications or exams are required, which tasks each qualification authorizes, and whether costs or training agreements apply if you leave.

No apprenticeship title, employer training program, course completion, or supervised task sign-off guarantees certification, responsible-charge authority, a permanent job, or transfer to another jurisdiction.

Free-first and low-cost learning

Start with the real local job market and certification rules before buying a private certificate:

1. Choose a target location and whether you want drinking-water, wastewater, collection/distribution, or a broader utility role.
2. Review 15–20 current postings and record repeated education, certification, shift, driving, physical, and experience requirements.
3. Identify the controlling operator-certification authority and facility class.
4. Search employer trainee roles, Apprenticeship.gov, public community or technical colleges, SENA or the relevant national institution, and employer-sponsored training.
5. Verify that a program is recognized for the exact certification path before paying.
6. Ask for total tuition, books, PPE, laboratory fees, exam fees, retests, travel, renewal, and refund terms in writing.
7. Prefer supervised, recognized, safety-controlled learning that connects to actual operator requirements.

Free online learning can support terminology, arithmetic, chemistry basics, microbiology basics, mechanical awareness, documentation, and interview preparation. It cannot replace plant-specific training, supervised practice, certification, qualified laboratory methods, confined-space roles, energy-control authorization, or legal operator requirements.

Funding pathways

In the United States, FAFSA may connect eligible students in eligible programs and institutions with federal grants, work-study, or loans. A short course, exam-preparation class, private certificate, or employer training program is not automatically aid-eligible.

Before borrowing, ask public colleges, apprenticeship sponsors, employers, workforce offices, veterans' education resources if applicable, community organizations, and program financial-aid offices about current support. Verify the exact program, eligibility, deadlines, repayment, tool or laboratory charges, travel, exam costs, and employment conditions in writing.

SENA and other public vocational systems may offer no-cost or lower-cost routes, but program availability and admission vary. No funding, seat, stipend, apprenticeship, certification, or job is guaranteed.

Wages and outlook

United States

BLS reports for **Water and Wastewater Treatment Plant and System Operators, SOC 51-8031**:

- **May 2024 median annual wage: \$58,260**
- **May 2024 median hourly wage: \$28.01**
- **Lowest 10 percent: below \$37,870**
- **Highest 10 percent: above \$86,160**
- **Employment in 2024: 132,400**
- **Projected employment in 2034: 123,800**
- **Projected change, 2024–2034: -8,700 jobs**

- **Projected change: -7 percent**
- **Projected openings: about 10,700 per year on average**, attributed by BLS to replacement needs rather than occupational growth

These are U.S. national occupational statistics, not starting-pay promises. The projected decline does not mean there will be no hiring; BLS still projects replacement openings. Pay and opportunity vary by location, employer, certification, plant type, shift, overtime, union status, and experience.

Canada

Government of Canada Job Bank reports national wage indicators for **NOC 92101**:

- **CAD \$25.00/hour low**
- **CAD \$36.06/hour median**
- **CAD \$48.00/hour high**

The wage page was updated November 19, 2025. These are occupational-group indicators, not guaranteed starting pay and not a currency-converted comparison with U.S. wages. Use regional filters and verify the actual facility, certification level, and duties.

No private salary estimate is included because current official wage evidence is available.

A practical 12-week starter plan

Weeks 1–2: Choose a target geography and compare drinking-water, wastewater, operator-in-training, and utility-operator postings. Record certification authority, plant class, education, shift, driving, and experience requirements.

Weeks 3–4: Study low-risk fundamentals: treatment vocabulary, arithmetic, chemistry and biology basics, pumps and valves conceptually, documentation, safety communication, and common utility terminology. Do not practice hazardous tasks.

Weeks 5–6: Compare operator-in-training routes, public technical/community-college programs, Registered Apprenticeship, employer training, SENA or the relevant national institution, and financial-aid eligibility. Verify recognition and total cost.

Weeks 7–8: Build supervised evidence in documentation, maintenance support, housekeeping, basic measurements, safe sample handling under instruction, computer use, customer or team communication, and reliable shift handover.

Weeks 9–10: Prepare a resume and interview examples showing accurate records, safe escalation, reliability, teamwork, careful observation, and respect for regulated scope. Describe only work you actually performed.

Weeks 11–12: Apply for trainee, operator-in-training, apprenticeship, utility-helper, treatment-operator, or related roles that match your actual eligibility. Track applications, certification steps, exams, deadlines, and follow-ups.

Resume and interview guidance

Useful evidence can include:

- accurate logs, inspections, inventory, work orders, or quality records;
- safe work around industrial or utility equipment;
- supervised sampling, laboratory support, maintenance support, or environmental work;
- arithmetic, measurement, chemistry, biology, mechanical, or electrical fundamentals;
- shift work, emergency communication, customer service, teamwork, or regulated documentation;
- computer systems, alarms, data entry, or controlled process-monitoring experience; and
- examples of stopping, escalating, or asking for qualified help when a condition was uncertain.

Do not claim certification, responsible-charge experience, SCADA authority, laboratory qualification, chemical-dosing authority, confined-space qualification, electrical authorization, lockout authorization, environmental compliance responsibility, or regulatory reporting authority unless it is true and supportable.

In interviews, ask:

- Which operator certificate or operator-in-training status is required at hire?
- What is the facility classification, and what certification level is needed for advancement?
- Which training, exams, and continuing education does the employer pay for?
- What does the first 90 days look like for a trainee?
- Which tasks require direct supervision or separate authorization?
- How are confined spaces, chemical hazards, biological exposures, energy control, rescue, and lone work managed?
- What shifts, callouts, overtime, weekends, and emergency duties are typical?
- How are laboratory work, SCADA access, process changes, and regulatory decisions separated by role?

Responsible use of AI

AI can help with low-risk career tasks such as:

- explaining general terminology in plain language;
- creating study questions from approved public material;
- organizing certification research and application deadlines;
- comparing job-posting requirements;
- improving resume structure without inventing experience; and
- drafting questions to ask a school, certification authority, or employer.

Do not use AI to:

- choose or calculate chemical doses or mixing instructions;

- interpret compliance samples or decide whether water or effluent is safe;
- change process controls, setpoints, alarms, or SCADA configurations;
- diagnose a plant hazard from incomplete information;
- generate or approve confined-space permits, lockout/tagout steps, bypasses, or emergency procedures;
- select PPE, atmospheric-monitor settings, rescue methods, or electrical test steps;
- override an alarm, approved procedure, laboratory method, permit, regulator, supervisor, or qualified operator; or
- make public-health, environmental-reporting, responsible-charge, or legal compliance decisions.

Treat AI output as unverified unless checked against current authoritative sources and the employer's approved information. Never paste credentials, facility diagrams, access details, process settings, chemical inventories, security information, customer records, laboratory results, incident details, or protected public-health/environmental data into an AI service unless the employer has explicitly approved that use.

Avoid scams and weak training offers

Be cautious when a provider promises guaranteed certification, guaranteed operator status, guaranteed employment, guaranteed wages, instant reciprocity, or a credential that supposedly works everywhere.

Before paying:

1. verify the provider and program directly with the controlling certification or education authority;
2. confirm whether the program counts toward the exact operator level and facility class you need;
3. ask for total cost, refund policy, exam fees, retests, books, laboratory charges, travel, renewal, and continuing-education costs;
4. verify current cohorts, placement claims, employer partners, and apprenticeship registration independently;
5. do not send money, identity documents, or banking information through an unverified recruiter or training link; and
6. keep copies of written promises and compare them with official rules.

A short private course may be useful, but it is not automatically recognized for state, provincial, territorial, Colombian, or other national operator requirements.

Before spending money

Use this order:

1. **Verify the job requirement.** Identify the location, facility type, facility class, and operator level.

2. **Verify the authority.** Check the current certification, education, experience, exam, renewal, and reciprocity rules.
3. **Check employer support.** Ask about paid training, exam reimbursement, apprenticeship, tuition assistance, and supervised progression.
4. **Check public options.** Compare community/technical colleges, SENA or the relevant national vocational institution, workforce programs, and eligible aid.
5. **Confirm recognition.** Get written confirmation that the exact program supports the credential you need.
6. **Calculate total cost.** Include tuition, fees, exams, retests, PPE, books, travel, renewal, and lost work time.
7. **Borrow last.** Never borrow based on a guarantee of licensing, employment, or salary.

Advancement

With experience, required certification, strong safety performance, and employer trust, a worker may progress through higher operator levels, lead or senior operator roles, specialized water or wastewater assignments, laboratory or maintenance pathways, collection/distribution work, training, supervision, utility operations, environmental support, or management.

Advancement depends on jurisdiction, plant classification, certification, experience, examination, continuing education, employer structure, and actual vacancies. A course or certificate does not guarantee promotion or responsible-charge authority.

Sources and verification links

United States

- U.S. Bureau of Labor Statistics, Occupational Outlook Handbook: <https://www.bls.gov/ooh/production/water-and-wastewater-treatment-plant-and-system-operators.htm>
- O*NET OnLine, 51-8031.00: <https://www.onetonline.org/link/summary/51-8031.00>
- U.S. EPA, Summary of State Operator Certification Programs: https://www.epa.gov/system/files/documents/2025-01/summary_of_state_operator_certification_programs.pdf
- OSHA, Hydrogen Sulfide in Workplaces: <https://www.osha.gov/hydrogen-sulfide/hydrogen-sulfide-workplaces>
- OSHA, Permit-Required Confined Spaces: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.146>
- OSHA, Hazard Communication: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200>
- OSHA, Bloodborne Pathogens: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1030>
- OSHA, Control of Hazardous Energy (Lockout/Tagout): <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.147>
- Apprenticeship.gov: <https://www.apprenticeship.gov/>

- Federal Student Aid, FAFSA student steps: <https://studentaid.gov/articles/fafsa-student-steps/>

Canada

- Government of Canada Job Bank, wages: <https://www.jobbank.gc.ca/marketreport/wages-occupation/20827/ca>
- Government of Canada Job Bank, requirements: <https://www.jobbank.gc.ca/marketreport/requirements/20827/ca>
- Health Canada, drinking water: <https://www.canada.ca/en/health-canada/services/environment/drinking-water.html>
- Environment and Climate Change Canada, wastewater: <https://www.canada.ca/en/environment-climate-change/services/wastewater.html>

Colombia

- SENA/Betowa, Operación de sistemas de potabilización de agua: <https://betowa.sena.edu.co/oferta/operacion-de-sistemas-de-potabilizacion-de-agua?level=2&modality=P&programId=106>
- SENA/Betowa, Saneamiento y salud ambiental: <https://betowa.sena.edu.co/oferta/saneamiento-y-salud-ambiental?modality=P&programId=167494>
- SENA/Betowa, Análisis de riesgos en sistemas de agua y saneamiento: <https://betowa.sena.edu.co/oferta/analisis-de-riesgos-en-sistemas-de-agua-y-saneamiento?location=57041298&modality=V&programId=173414>
- Ministry of Housing, RAS Title C: <https://ventanilla.minvivienda.gov.co/sites/default/files/documentos/titulo-c-dic-4-2013.pdf>
- Resolution 2115 of 2007: <https://ventanilla.minvivienda.gov.co/sites/default/files/normativa/2115%20-%202007.pdf>

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- OIT/Cinterfor institutional network: <https://www.oitcinterfor.org/red-institucional>
- OIT/Cinterfor REDIFP resources: <https://www.oitcinterfor.org/recursos/redIFP>

Limitations and non-guarantees

This guide does not guarantee employment, wages, openings, admission, funding, apprenticeship, certification, licensing, reciprocity, authorization, promotion, or earnings.

Requirements can change. Verify current rules with the controlling certification, drinking-water, wastewater, environmental, labor/safety, education, accreditation, and employer authorities before spending money or relying on a credential.

This guide does not claim independent human review, professional translation certification, accessibility certification, legal review, environmental compliance review, laboratory approval, engineering approval, electrical authorization, industrial-hygiene review, medical review,

confined-space or rescue qualification, operator licensing determination, accreditation, or financial advice.